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STEP 3 - Layer by layer: where does the context get 'written into' the vector?
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layer 0: same-sense 0.958 | different-sense 0.956 | gap +0.002
layer 1: same-sense 0.876 | different-sense 0.761 | gap +0.115
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# -----
banner("STEP 3 - Layer by layer: where does the context get 'written into' the vector?")
# -----
within, across = [], []
for L in range(n_layers + 1):
    V = F.normalize(vecs[:, L, :], dim=-1)
    S = (V @ V.T).numpy()
    w = np.concatenate([S[np.ix_(river, river)][np.triu_indices(river.sum(), 1)],
                        S[np.ix_(~river, ~river)][np.triu_indices((~river).sum(), 1)]]
    within.append(w.mean())
    across.append(S[np.ix_(river, ~river)].mean())
    print(f" layer {L:2d}: same-sense {within[-1]:.3f} | different-sense {across[-1]:.3f} "
          f"| gap {within[-1] - across[-1]:+.3f}")

plt.figure(figsize=(8, 4.5))
plt.plot(within, "o-", label="same sense (river-river, money-money)")
plt.plot(across, "s-", label="different sense (river-money)")
plt.fill_between(range(n_layers + 1), across, within, alpha=0.15)
plt.xlabel("layer (0 = input embeddings)")
plt.ylabel("mean cosine similarity")
plt.title(f"'{TARGET}': the gap between senses grows with depth")
plt.legend()
plt.grid(alpha=0.3)
show("ex1_step3_layers")
```